

How ANTI-COLLISION System Works For Trains



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An anti-collision system for trains is an electronic system that is designated to ensure safety of the passengers and proper operation of trains. It helps avoid the risk of dangerous and unexpected collisions of trains, which may be caused due to human errors or equipment failures. The GPS system described here was designed and invented by Rajaram Bojji and patented by Konkan Railway Corp. Ltd.

Anti-collision devices (ACDs) rely on global positioning system (GPS) satellites for position updates. These exchange information through radio frequency transmissions to automatically brake and prevent collisions. Applications of such ACDs have been refined to not only

prevent midsection collisions but also to prevent their occurrences in station yards.

Infrared ACD

Let us take an example. An infrared (IR) sensor and IR transmitter is fitted on the locomotives of trains 1 and 2. The system works in the line-of-sight. The IR sensor is connected to the braking circuit on each train, which is further connected to a stepper motor that applies brakes to the train whenever it senses an approaching train in its line-of-sight.

Drawbacks of such a system are:

1. The system fails on a curved path as it only works in line-of-sight
2. Cannot prevent rear-end, side-track collisions
3. Range of the IR system depends on IR transmitter power, which is inversely proportional to reaction time; the more the power, the lesser the reaction time
4. The system is affected by weather conditions such as rain and storms that interfere with the IR signal

GPS ACD

A GPS based ACD is a self-acting micro-processor based electronic device. It takes inputs from GPS satellites and networks with other ACDs to detect collision-like situations. It has a range of about three kilometres for inter-ACD communication.

A GPS based ACD has four components:

1. GPS for getting position updates of locomotives
2. Command and control unit (CCU), which is the decision-making device in the system
3. Radio trans-receiver for inter anti-collision device radio communication
4. Automatic braking unit for stopping the locomotive

ACD units

Locomotive ACD (fixed and moving).
It is installed in the locomotive perma-

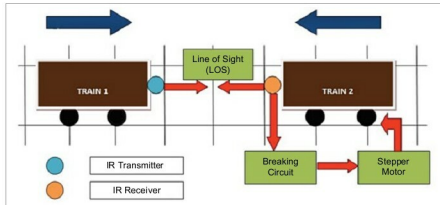


Fig. 1: An IR based ACD system

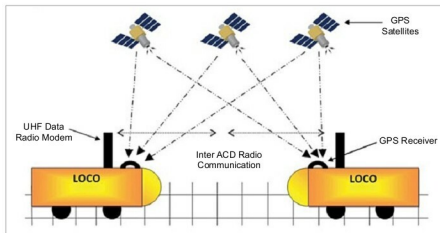


Fig. 2: A GPS based ACD system